FG Supply Planning as AI Labor

Agent-Owned Decisions, Mathematical Grounding, and UX Primitives

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1. Where FG Supply Sits in the Planning Hierarchy

Daybreak's AI-as-Labor operating model has established two agent-owned decisions: the **Unconstrained Forecast (UCF)** owned by the Demand Planning Agent, and the **Allocation Commit (AC)** owned by the Allocation Planning Agent. Between these sits a decision layer that neither agent owns today: **how much finished goods supply to bring into the network, from which sources, and when.**

This is the FG Supply Planning problem. It consumes the UCF as its primary demand signal and produces the supply position that the Allocation Agent takes as a constraint. Without an agent-owned supply decision, the planning hierarchy has a gap: demand judgment compounds, allocation judgment compounds, but the supply decisions that connect them remain trapped in manual ERP workflows, planner spreadsheets, and ad-hoc MRP runs that discard decision lineage every cycle.

The planning cascade flows as follows:

```
Demand Planning Agent → UCF (unconstrained demand signal)
        ↓
FG Supply Planning Agent → Supply Commit (what to buy/make, when, from whom)
        ↓
Allocation Planning Agent → Allocation Commit (who gets what, when)
        ↓
Execution (order promising, fulfillment)
```

Each layer owns a scoped decision. Each layer receives grounding from mathematical baselines. Each layer is governed by policy and risk thresholds, not universal review. The FG Supply Planning Agent fills the structural gap.

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2. The AI-Owned Decision: Supply Commit (SC)

Definition

The **Supply Commit (SC)** is the agent-owned decision artifact for the FG Supply Planning use case.

The Supply Commit is defined as: the best **demand-informed, constraint-respecting** set of supply actions (purchase orders, production orders, transfer orders) across items, sources, locations, and time buckets that balances inventory investment against service risk within the policy envelope.

Minimum decision artifact:

- `order_qty [item, source, destination, time_bucket]` — what to buy/make/transfer, how much, when
- `target_inventory_position [item, location, time_bucket]` — the intended inventory trajectory
- `projected_otif [item, segment, time_bucket]` — expected service outcome given this supply plan
- `projected_inventory_cost [item, location, time_bucket]` — carrying cost, obsolescence exposure, expedite premium
- `supply_pegging [demand_signal → supply_action]` — which demand drove which supply decision and why

What the FG Supply Planning Agent Owns

- Applying approved policies, constraints, and demand signals to produce the Supply Commit
- Explaining supply decisions through demand pegging and constraint attribution
- Detecting when the decision is unsafe to auto-submit under policy
- Recommending source selection across eligible suppliers based on cost, lead time, reliability, and capacity

Explicit Ownership Boundary

The FG Supply Planning Agent does **not** own:

- The Unconstrained Forecast (owned by Demand Planning Agent)
- Supplier negotiation, contract terms, pricing agreements, or sourcing strategy
- Capacity expansion decisions, capex commitments, or make-vs-buy strategy
- Allocation of constrained supply to customer segments (owned by Allocation Planning Agent)
- S&OP or IBP consensus outputs and cross-functional reconciliation
- Financial targets, budgets, or working capital limits as decision logic (these are policy constraints, not agent objectives)

Governance Model for the SC

Human supervision of the Supply Commit is permitted only to:

- Correct missing or wrong grounding inputs (lead times, MOQs, supplier capacity, costs, inventory positions)

- Inject documented exception context not available to the agent (supplier disruptions, port closures, quality holds, strategic forward buys approved by leadership)
- Tune policy and risk thresholds

Human intervention may **not**:

- Convert supply planning into a consensus workflow where negotiation replaces judgment
- Introduce allocation logic or customer-segment prioritization into supply decisions (this belongs to the Allocation Agent)
- Repeatedly override the same decision identity without addressing upstream data or policy gaps

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3. Mathematical Grounding: The Supply Baseline Pack (SupBP)

Design Principle

Following the established pattern — FBP grounds the Demand Agent, SBP grounds the Allocation Agent — the **Supply Baseline Pack (SupBP)** provides mathematical grounding for the FG Supply Planning Agent. It is non-authoritative. It cannot be published as the Supply Commit. It is evidence the agent may accept, reject, or blend.

SupBP Contents

1) Candidate Supply Plans (one or more)

Each candidate is a complete supply plan at the decision granularity.

- `candidate_id` (e.g., `MRP_STANDARD_V1`, `SAFETY_STOCK_OPTIMIZED_V1`, `MIN_COST_LP_V1`, `SERVICE_MAXIMIZED_V1`, `PARAMETRIC_CFA_V1`)
- `order_qty [item, source, destination, time_bucket]`
- `target_inventory_position [item, location, time_bucket]`
- `projected_otif [item, segment, time_bucket]` (service outcome implied by this candidate)
- `projected_cost [item, location, time_bucket]` (inventory carrying + ordering + expedite)
- `projected_excess [item, location, time_bucket]` (obsolescence/overstock exposure)

Why multiple candidates matter: Supply planning involves an explicit tradeoff frontier between service (OTIF) and investment (inventory cost). A single MRP output hides this tradeoff. Multiple candidates expose it — one candidate may maximize service at high cost, another may minimize cost at lower service, a third may represent the Pareto-efficient compromise. The agent reasons across candidates; the human supervises the tradeoff, not the arithmetic.

2) Demand grounding (consumed from UCF)

- `ucf_hash` — the specific UCF version used as input
- `demand_by_item_location_bucket` — the demand signal driving the supply plan
- `demand_uncertainty [quantiles or prediction intervals]` — consumed from FBP via UCF
- `demand_classification [item]` — smooth, intermittent, lumpy, new (affects method selection)

3) Supply-side constraint diagnostics

These enable the agent to explain "why this order quantity" and detect unsafe outputs.

- `supplier_capacity [source, item, time_bucket]` — available capacity by supplier
- `lead_time_distribution [source, item]` — expected and variance, not just point estimate
- `moq_and_lot_size [source, item]` — minimum order quantities and increment constraints
- `supplier_reliability_score [source]` — historical on-time-in-full performance
- `cost_structure [source, item]` — unit cost, freight, volume discount tiers, payment terms
- `open_orders [item, source, destination, expected_receipt_date]` — in-transit and on-order supply
- `substitution_map [item → eligible_alternates]` — where substitution is permitted

4) Inventory position and health diagnostics

- `on_hand [item, location]`
- `in_transit [item, location, expected_date]`
- `committed_to_orders [item, location]` — already allocated or promised
- `available_to_plan [item, location]` — net available for new supply decisions
- `days_of_supply [item, location]` — current coverage vs demand rate
- `excess_and_obsolescence_exposure [item, location]` — projected E&O at current trajectory
- `shelf_life_remaining [item, location, lot]` — where applicable

5) Tradeoff surface and sensitivity signals

The supply planning equivalent of shadow prices — "where does one more unit or one more day of lead time matter most?"

- `marginal_service_impact [item, location]` — projected OTIF change from ±1 unit of safety stock
- `marginal_cost_impact [item, source]` — cost delta from shifting one unit between sources
- `lead_time_sensitivity [item, source]` — OTIF impact if lead time increases by 1 day
- `stockout_probability [item, location, time_bucket]` — given current plan and demand uncertainty

- cost_to_serve_by_source [item, source] — fully loaded cost comparison across sources

6) Versioning and reproducibility

- supbp_version_hash
- ucf_hash (demand input traceability)
- constraint_set_hash
- inventory_snapshot_hash
- solve_timestamp
- solver_config [method, parameters, objective_weights]

Governance-safe rule: Every published Supply Commit must reference the SupBP hash used at decision time to preserve decision identity and enable outcome attribution.

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4. Policies and Risk Thresholds

Policies do not modify the Supply Commit. They only control autonomy and review priority.

Policy Class: Supply Plan Integrity

These policies ensure the Supply Commit is internally coherent and executable.

Policy	Purpose	Trigger Condition	Governance Action
Inventory Balance Policy	Prevents plans that violate inventory physics	Projected inventory goes negative in any bucket	**Suggested + Block submission**
Lead Time Feasibility Policy	Prevents orders that cannot arrive in time	Order receipt date precedes source lead time floor	**Suggested + Block submission**
MOQ/Lot Size Compliance Policy	Ensures orders respect supplier constraints	Order quantity violates MOQ or lot-size increment	**Suggested**
Demand Coverage Policy	Ensures supply plan addresses the demand signal	Material demand periods have no supply coverage and no documented reason	**Suggested**

Policy Class: Supply Outcome Risk

These policies govern whether the Supply Commit is safe to auto-submit given business exposure.

Policy	Purpose	Trigger Condition	Governance Action
Service Risk Policy	Prevents unacceptable OTIF degradation	Projected OTIF for any governed segment falls below floor	**Suggested**
Inventory Investment Policy	Prevents excess inventory accumulation	Projected days-of-supply exceeds ceiling for item/location	**Suggested**
Supplier Concentration Policy	Prevents over-reliance on a single source	Single-source share exceeds threshold for governed items	**Suggested**
Expedite Exposure Policy	Flags abnormal premium freight or rush orders	Expedite cost in the plan exceeds threshold	**Suggested**
E&O Exposure Policy	Prevents avoidable write-offs	Supply plan increases projected excess/obsolete beyond threshold	**Suggested**
Plan Stability Policy	Prevents unexplained whiplash in orders	Period-over-period order quantity change exceeds tolerance	**Suggested**

Policy Enforcement Rules

- Policy violations never change the Supply Commit values.
- Policy violations only affect review status (Autonomous vs. Suggested) and, for integrity violations, submission eligibility.
- Multiple violations increase review priority, not authority.
- No policy may introduce allocation logic, customer prioritization, or financial targets as decision constraints.

Risk Dimensions

- **Service Exposure:** Expected OTIF degradation on critical items/segments
- **Investment Exposure:** Excess inventory carrying cost or working capital impact
- **Expedite Cost Exposure:** Premium freight, overtime, rush surcharges
- **Obsolescence Exposure:** Projected write-offs from overstocking perishable or end-of-life items
- **Supply Continuity Exposure:** Risk of supplier disruption on concentrated sources
- **Revenue Exposure:** Expected revenue loss from stockouts on high-velocity items

5. Supervision Loop and Decision Status Paths

Supervision Loop (Not an Approval Workflow)

1. **Observe:** Review supply decisions outside policy, prioritized by business impact.
2. **Orient:**
    - Compare agent Supply Commit vs. SupBP candidates (e.g., did the agent deviate from the MRP baseline? Why?)
    - Inspect the service-vs-investment tradeoff surface — is the agent's position on the Pareto frontier defensible?
    - Review demand pegging: which demand signals drove the largest supply actions?
    - Check supplier reliability signals: is the agent buffering appropriately for unreliable sources?
    - Inspect stockout probabilities and lead time sensitivities for flagged items
3. **Decide:** Accept or override with reason, confidence, and evidence.
4. **Act:** Commit the Supply Commit for downstream consumption (by the Allocation Agent and execution systems).

Decision Status Paths

```
Suggested → Reviewed → Accepted → Submitted
Suggested → Reviewed → Overridden → Submitted
Autonomous → Reviewed → Accepted → Submitted
Autonomous → Reviewed → Overridden → Submitted
Autonomous → Not Reviewed → Auto-Submitted
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Policy Logic

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Suggested =
  (Policy check(s) fails AND $ impact > risk threshold)
  OR (Supply Commit delta vs prior cycle > risk threshold)
  OR (Supply integrity violation is detected)
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6. UX Primitives for the Supply Planner

The interaction model follows the same six UX primitives established for Demand and Allocation, with scope and decision type adapted to the supply planner persona.

Primitive	Supply Planner Experience
Scheduled Tasks & Digests	"Pipeline ran: 847 supply decisions committed autonomously. 23 need your attention before the 2pm PO cutoff. Top risk: nitrile gloves projected at 88% OTIF vs 96% floor — Supplier A lead time drift detected."
Worklist	Exception triage by policy violation type. Sorted by $ exposure. Each item shows: agent recommendation, SupBP candidate comparison, projected OTIF, projected inventory position, and gating constraint.
Chat	"Why is the agent ordering 40% more from Supplier B this week?" → Agent cites Supplier A reliability score drop (from 91% to 78% over 4 weeks), lead time sensitivity analysis showing 3-point OTIF risk, and the cost-to-serve delta between sources. Planner can inject context: "Supplier A warned us about port delays through March — extend buffer through Q1."
Task Log	Audit trail: which SupBP candidates were generated, which the agent selected/blended, what policies were evaluated, which decisions were auto-submitted.
Agent Configuration	Policy thresholds (OTIF floors, DOS ceilings, expedite caps), supplier parameters (reliability weights, lead time buffers), demand classification overrides, source eligibility rules. Initially Daybreak-configured, customer-maintained.
Dashboards	Inventory health by location, supplier performance trends, service-vs-investment Pareto visualization, projected stockout heatmap, E&O exposure waterfall. Structured analysis surfaces — not the primary workflow.

The Chat Interaction Model for Supply

The supply planner's chat interactions follow a pattern distinct from demand planning. Where the demand planner asks "why is this item flagged?" the supply planner asks tradeoff questions:

- *"What happens to OTIF if I delay this PO by one week?"* → Agent runs sensitivity from SupBP, shows projected OTIF impact by segment, flags if Allocation Agent would need to re-prioritize.
- *"Show me the cost difference between fulfilling this demand from Supplier A vs. Supplier B."* → Agent cites cost-to-serve including freight, lead time risk premium, and MOQ implications.
- *"Supplier C just told me they're allocating us 30% less next month. Update the plan."* → Agent ingests the constraint, re-solves supply plan, shows downstream impact on OTIF and inventory, and flags items that may need the Allocation Agent to re-run.
- *"Can we substitute SKU 4823 with SKU 4825 for the clinic segment?"* → Agent checks substitution eligibility, recalculates supply coverage, and shows net service and cost impact.

Every chat interaction that changes the Supply Commit captures the planner's reasoning as structured override context — tied to the decision identity, scored against outcomes in subsequent cycles, absorbed or rejected based on value added.

Key Metrics

Metric	Definition
Touchless Rate	% of Supply Commits not reviewed by a human
Agent DQS	Outcome lift attributable to agent Supply Commits (OTIF achieved vs. baseline, inventory cost vs. baseline)
User DQS	Outcome lift from human overrides vs. the agent baseline
Cognitive Load Index	Decisions reviewed ÷ total decisions in scope
Override Dependency Ratio (Kill Metric)	Repeated human intervention on the same decision identity across cycles
Downstream Coherence	Divergence between Supply Commit and what the Allocation Agent actually receives as available supply (proxy for integration quality)

If the Supply Commit requires persistent human intervention to produce acceptable service and inventory outcomes, the system is **not** AI-as-Labor. It is a recommendation engine with extra steps.

Summary: How This Fits the Thesis

The FG Supply Planning Agent completes the planning cascade. Demand judgment compounds in the UCF. Supply judgment compounds in the SC. Allocation judgment compounds in the AC. Each agent owns a scoped, repeatable decision. Each is grounded by a versioned mathematical baseline pack. Each is governed by policies and risk thresholds, not universal review. Each captures overrides with context, scores them against outcomes, and absorbs or rejects them.

The Supply Commit is not a forecast. It is not an allocation. It is the decision that converts demand intelligence into supply action — and it is the decision that most directly determines whether the distributor hits its OTIF targets without drowning in inventory. Today, this decision resets every MRP cycle. With Daybreak, it compounds.